

Examining ethnic-racial identity development and adjustment among Colombian adolescents

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ABSTRACT

The current study examined ethnic-racial identity (ERI) development and adjustment among Colombian adolescents ($N = 462$, $M_{\text{age}} = 15.90$, $SD = 1.15$, 47% female) from six schools in Medellín, Colombia. Findings from chain sequential mediation analyses with bias-corrected bootstrapping indicated that ERI exploration was positively associated with ERI resolution ($\beta = 0.48$), which was positively associated with global identity cohesion ($\beta = 0.29$); in turn, higher global identity cohesion was associated with higher academic engagement ($\beta = 0.26$), higher self-esteem ($\beta = 0.50$), lower GPA ($\beta = -0.12$), and lower depressive symptoms ($\beta = -0.35$). ERI exploration ($\beta = 0.14$) and resolution ($\beta = 0.11$) were each positively associated with other group orientation. Findings provide preliminary evidence for the promotive role of ERI among Colombian adolescents, supporting a conceptualization of ERI as a developmental competency.

Adolescence is a critical period during the lifespan for developing an understanding of oneself and forming an identity (Erikson, 1968). Individuals' identities are multifaceted and informed by various social identities (Crogetti, 2017). Self-categorization theory suggests that in societies where ethnoracial¹ inequities are pervasive, one's social identity based on ethnicity-race will be salient (Turner et al., 1987). In such societies, ethnic-racial identity (ERI) development, which involves the processes of exploring and developing clarity regarding one's ethnic heritage and membership in a socially constructed racial group (Umaña-Taylor et al., 2018), becomes a relevant construct and developmental task for young people. Furthermore, Erikson's (1968) theory of psychosocial development suggests that exploration and resolution of identity domains should lead to a more integrated sense of self (i.e., global identity cohesion), which promotes positive psychosocial adjustment. These theoretical ideas have been examined with ethnoracially diverse samples of adolescents in the United States and empirical evidence has supported the notion of ERI as a developmental

competency (Umaña-Taylor, 2018). Furthermore, randomized controlled trials that have tested the efficacy of ERI-focused interventions in the United States (Umaña-Taylor et al., 2018) and in Europe (Ceccon et al., 2023; Juang et al., 2020; Schachner et al., 2024) have provided evidence to support the feasibility of intervening in this developmental competency in a manner that promotes positive youth outcomes – underscoring the important practical implications of this work for adolescents' lives. A significant gap in the literature, however, is our limited understanding of these processes with youth in Latin American countries such as Colombia (e.g., Schwartz et al., 2012), despite a robust literature documenting that marginalization and oppression as a function of race and ethnicity is prevalent in Colombia (e.g., Castillo and Carou, 2002; Chavez-Dueñas et al., 2014; Hristov, 2009; Perreira and Telles, 2014). As such, the current study was designed to examine these theoretical notions among a sample of Colombian adolescents and begin to build an empirical evidence base to support the development of future intervention efforts.

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¹ Consistent with Umaña-Taylor et al. (2024), we use the term ethnoracial “to acknowledge the intersecting nature of ethnicity and race in the psychological process of forming an identity based on individuals' sometimes-overlapping lived experiences that are shaped by their ethnic and racial group memberships...and acknowledge the experiences of individuals belonging to social identity groups that have been socially constructed as ‘ethnic’ (i.e., Latinx/Hispanic; Humes et al., 2011), but whose lived experiences reflect a racialized othering effect (Massey, 2014).”

ERI theory and associations with adjustment

In societies characterized by ethnoracial inequities that make social identities based on race and ethnicity salient, identity development includes adolescents considering the role that their ethnoracial group membership plays in their lives (Verkuyten, 2016). ERI development includes the processes of exploring and gaining clarity (i.e., resolution) regarding one's ethnic heritage and membership in a socially constructed racial group (Umaña-Taylor, 2018). Individuals are in a constant quest for identity synthesis, in which they integrate their multiple social identities along with their thoughts about their past, their present, and their future, to come to a sense of clarity regarding who they are and who they hope to become (Erikson, 1968). Developmental theory further suggests that via a process of in-depth exploration regarding one's identity, individuals gain clarity and a sense of understanding (i.e., resolution) regarding their self-concept that contributes to building a more integrated sense of self (i.e., global identity cohesion); in turn, this sense of global identity cohesion promotes positive psychosocial adjustment (Erikson, 1968; Syed and McLean, 2016).

Better psychosocial adjustment is expected when these developmental processes are engaged because of the increased clarity and confidence that individuals gain through exploration, consideration of multiple options for their identities, and attaining a clearer and well-informed understanding of themselves (Umaña-Taylor, 2018). These theoretical notions have been applied to the study of ERI in the U.S. For instance, empirical evidence with African American participants has demonstrated that exhibiting high ERI exploration without ERI resolution (i.e., moratorium ERI status) is more likely among younger adolescents who are less advanced in their ERI development than among college students and adults; furthermore, these older participants, who are further along in their ERI development, are more likely to exhibit simultaneously high levels of exploration and resolution (i.e., Achieved ERI status) relative to adolescents (Yip et al., 2006). Similarly, with a sample of U.S. Latino adolescents who were followed from 9th to 12th grade, the most common trajectory of ERI change during high school was one characterized by increasingly higher exploration and resolution over time (Douglass and Umaña-Taylor, 2015a). Moreover, higher ERI exploration and resolution has been significantly associated with higher global identity synthesis and lower global identity confusion (Safa et al., 2024). Finally, experimental evidence has supported this theoretical progression of increases in exploration leading to increases in resolution, which in turn promotes greater identity cohesion and adjustment in the form of higher self-esteem, lower depressive symptoms, and better academic grades (Umaña-Taylor et al., 2018). This has been referred to as a developmental cascade model, given that engaging in the developmental process of ERI exploration promotes ERI resolution, and this increased resolution that emerges as a function of in-depth exploration promotes another developmental need: global identity cohesion; in turn, the psychosocial benefits of global identity cohesion for adolescents promote better adjustment during adolescence (Umaña-Taylor, 2018).

Conceptually, engaging in exploration provides young people with increased knowledge about their ethnoracial background and an actively informed understanding of who they are in this regard, which they have gained through a process of self-exploration rather than an unconscious adoption of others' views of how they should identify (Umaña-Taylor, 2016). This active and intentional engagement in self-discovery related to one's ERI is expected to provide youth with a more mature sense of self, increased self-confidence regarding how they view themselves with respect to their ERI, and tools that can enable them to understand how to cope with negative ERI-related experiences (Umaña-Taylor, 2016). Furthermore, many psychosocial benefits (e.g., high self-esteem) are theorized to emerge from this more mature understanding and greater sense of self-confidence that ultimately supports young people as they navigate challenging ethnoracially based experiences (Umaña-Taylor, 2018).

These conceptual notions in relation to the construct of ERI have

been extensively studied in the U.S. and there is an increasing body of literature examining this construct (referred to as ethnic or cultural identity) in Western European countries as scholars examine the minoritized nature of the identities of immigrants and people from non-majority ethnic or religious groups (e.g., Jugert et al., 2020; Verkuyten, 2016). A large body of empirical work has emerged and demonstrated that the developmental processes of ERI exploration and resolution are positively associated with indices of adolescent adjustment and negatively associated with indices of maladjustment for ethnoracial minoritized youth in the U.S. (Umaña-Taylor and Rivas-Drake, 2021). Emerging research in Latin America has found evidence of similar promotive effects among ethnoracially diverse Costa Rican adolescents (Castro, 2002) and minoritized youth in Chile and México (Briceño-Olivera et al., 2020; Guitart et al., 2011). Yet, it is unknown whether similar associations would emerge among adolescents in Colombia. Examining these associations among adolescents growing up in Colombia can contribute to our understanding of the universality and cultural specificity of theoretical notions suggesting that ERI is an important developmental competency during adolescence. Furthermore, as such evidence builds, researchers can explore possibilities for intervention in ERI with Colombian adolescents, as has already been done in numerous European countries (see Juang et al., 2023, for an overview).

In terms of specific outcomes of interest in the current study, we examined indices of adjustment focused on mental health (i.e., depressive symptoms), general self-concept (self-esteem), academic adjustment (i.e., school grades, academic engagement), and attitudes toward engaging with others who are ethnoracially different from oneself (i.e., other group orientation). Prior theoretical and empirical work has provided strong support for significant associations between ERI and each of these indices of adjustment, noting the personal psychosocial and relationship-based benefits that accompany greater identity synthesis and less confusion (e.g., Umaña-Taylor, 2018). Specific to the Colombian context, prior research has identified psychological (e.g., Gómez-Restrepo et al., 2023) and academic (e.g., Gamboa and Londoño, 2015) indicators of adjustment as important areas of national concern for Colombian youth.

ERI as a salient developmental construct in Colombia

Ethnoracial inequities exist in many societies around the world, and what distinguishes these societies is the specific domain (e.g., race, ethno-religious heritage, migration status) that is used to grant access to power and resources, as well as the extent to which a given society acknowledges the source of these inequities as based on race or ethnicity. Like many countries in Latin America that experienced the Spanish conquest, Colombia is characterized by a history of deep-seated racism and a distribution of power and resources based on an ethnoracial hierarchy that favored those who were racially White and of European descent and oppressed those of Indigenous or African descent (Castillo and Carou, 2002; Chavez-Dueñas et al., 2014). Furthermore, constitutional policies that emphasized *mestizaje* (i.e., everyone is assumed to be racially mixed) facilitated the ability for those in power to deny the existence of racism and enabled an argument (politically and scholarly) that experiences of marginalization and inequality were exclusively a function of social class. As a result, scholarly discourse on race or ethnicity as salient domains of individuals' identities has been essentially absent.

Despite this absence in the literature, an examination of Colombia's social and political landscape suggests this is an important area on which to focus when considering adolescents' development and adjustment. In an effort to enhance their national-democratic legitimacy during the transition out of political authoritarianism and repression, national elites in many Latin American countries pushed for and passed multicultural citizenship reforms (including a new Colombian Constitution in 1991) that established rights (e.g., property rights, official status for

minority languages, guarantees of bilingual education) for groups that historically had been marginalized, such as those of Indigenous or African descent (Hooker, 2005). In Colombia there is evidence with adults that the reforms resulted in an increased salience of collective group identities for traditionally marginalized groups because a requisite for access to these resources is that groups prove the distinctiveness of their heritage culture (relative to mainstream Colombian culture) and their connection to an ancestral territory (Chaves and Zambrano, 2006). Thus, one result of multiculturalism reforms was ethnic *re-socialization* in which members of marginalized groups were motivated to learn the history and ancient ancestral practices of their culture to “prove” their historically marginalized status (Castillo and Carou, 2002; Chaves and Zambrano, 2006).

Moreover, in a parallel line of work not focused on youth identity, social behavioral scientists documented the pervasive nature of ethnoracial discrimination in Colombia (e.g., Hristov, 2009), and the detrimental consequences of historical discrimination on health problems, low literacy, formal education, and high poverty (Chavez-Dueñas et al., 2014). Furthermore, a study that explored variability within Colombians demonstrated the pervasive nature of discrimination in Colombia based on skin color and phenotype, even after controlling for socioeconomic class (Perreira and Telles, 2014). Finally, in one study that addressed issues of ethnoracial discrimination with youth, Colombian adolescents demonstrated a high level of awareness and concern about social discrimination (including ethnoracial; Frydenberg et al., 2001). Thus, the existing literature suggests that ERI may indeed be salient to Colombian adolescents and worthy of examination.

The current study

In the U.S., race and ethnicity are salient social identities due to a historical context in which access to socio-political power and resources have been distributed in a manner that disenfranchises ethnoracial minoritized groups (Omí and Winant, 2014). This broader context has made ERI formation a salient feature of development for youth. Given Colombia’s sociopolitical history in which Indigenous and Afro-descendant youth have been disadvantaged with respect to economic, social, and educational opportunities (Preiss et al., 2015), similar processes may emerge. Accordingly, the current study examined a developmental cascade model in which ERI exploration was hypothesized to be positively associated with ERI resolution, which in turn was hypothesized to be positively associated with global identity cohesion; in turn, global identity cohesion was hypothesized to be negatively associated with depressive symptoms and positively associated with adolescents’ grades, academic engagement, self-esteem, and other group orientation.

Method

Participants

Participants represented a community-based convenience sample recruited during 2017–2018 from six schools in Medellín, Colombia, a major metropolitan city with a population of over 3.9 million people (DANE, 2018). Schools were selected to reflect the socioeconomic diversity of the country. Two schools represented the two lowest socioeconomic status (SES) groups, two represented the middle SES groups, and two represented the highest SES groups.² Participants were 462

adolescents ($M_{age} = 15.90$, $SD = 1.15$ years; 47.3% female) in Grades 9 (36.4%), 10 (31.8%), and 11 (31.4%), which corresponds to the last three years of secondary education in Colombia. Participants’ self-identified ethnoracial background included *Mestizo* (i.e., mixed Indigenous and European ancestry; 39.2%); *Latino* (i.e., transnational label used by individuals of Latin American descent; 32.3%); *Blanco* (i.e., White; 16.5%); *mezcla de Blanco y Negro* (i.e., mix of White and Black; 4.1%); *mezcla de Blanco e Indígena* (i.e., mix of White and Indigenous; 1.5%); *Afrodescendiente* (i.e., African-descent; 1.3%); *Negro* (i.e., Black; 1.1%); or another designation (4.0%; e.g., *Indígena*; *Mestizo y Latino*; *Blanco y Latino*). Most participants were born in Colombia (94.8%), and those born in other countries (e.g., Venezuela, United States, México, Chile, Spain) had resided in Colombia for an average of 7.50 years. Based on the government-assigned SES designation² provided by youths’ parents/guardians, 6.9% of participants were considered low-low SES, 26.8% were low SES, 28.8% upper-low SES, 13.0% medium SES, 13.6% medium-high SES, 7.4% high SES, and 3.5% did not report SES.

Procedures

Participants who provided parental consent and youth assent completed paper surveys during school time, which took approximately 45 min. All materials were provided in Spanish. Measures or demographic questions not available in Spanish were translated into Spanish and back-translated into English by separate individuals. Discrepancies were discussed by the research team, which included a native Spanish-English bilingual Colombian-U.S. American faculty member, two native Spanish-speaking Colombian faculty members with English proficiency, and four native Spanish-speaking Colombian Psychology students with English proficiency. Modifications to Spanish items were made as needed, prioritizing meaningful, rather than literal, translation. A final review of the entire survey was conducted by two Colombian members of the research team with a focus on clarity in the context of Colombian dialect. Research team members distributed surveys, monitored data collection, and answered questions.

Measures

ERI exploration and resolution

Adolescents were asked to report on their ERI exploration (3 items; e.g., “I have attended events that have helped me learn more about my ethnicity;” $\alpha = 0.71$) and ERI resolution (3 items; e.g., “I know what my ethnicity means to me;” $\alpha = 0.83$) using respective subscales from the Ethnic Identity Scale – Brief (Douglass and Umaña-Taylor, 2015b). Instructions stated: “The next statements refer to ‘ethnic identity,’ or how much one feels they identify with the characteristics of an ethnic or racial group.” Responses ranged from *Does not describe me at all* (1) to *describes me very well* (4). Higher scores reflected higher levels of ERI. The subscales have demonstrated adequate reliability and support for construct validity with U.S. ethnoracially diverse youth (Umaña-Taylor et al., 2018) and youth in Latin America (Briceño-Olivera et al., 2020).

Global identity cohesion

Adolescents’ global identity cohesion was assessed using a modified version of the 12-item Erikson Psychosocial Stage Inventory (Rosenthal et al., 1981) with two subscales, personal identity synthesis (e.g., “I’ve got a clear idea of what I want to be”) and confusion (e.g., “I don’t really know who I am”). Responses ranged from *strongly disagree* (1) to *strongly agree* (5). Higher scores reflected greater global identity cohesion ($\alpha = 0.76$). The scale has demonstrated adequate reliability and validity with U.S. ethnoracially diverse youth (Umaña-Taylor et al., 2018) and results from psychometric testing have provided support for the use of this measure in Spanish (Schwartz et al., 2014).

² Socioeconomic status (SES) is assigned by the national government based on housing characteristics and living conditions (e.g., type of housing structure, housing materials, degree of overcrowding per unit). SES labels used by the Colombian government are (1) *bajo-bajo* [low-low], (2) *bajo* [low], (3) *medio-bajo* [middle-low], (4) *medio* [middle], (5) *medio-alto* [middle-high], (6) *alto* [high] (DANE, 2021).

Grade point average (GPA)

Adolescents' GPA was calculated based on their quantitative grades obtained from end-of-school-year official school transcripts for the 2017–2018 academic year in core courses that were required for each grade level. Students' grades ranged from 1.0 to 5.0, with 5 indicating superior performance.

Academic engagement

Adolescents' level of engagement in school was assessed using the 10-item Academic Engagement Measure (Skinner et al., 2008), which assesses behavioral (e.g., "I try hard to do well in school") and emotional (e.g., "When I am in class, I feel good") engagement. Responses ranged from *never* (0) to *all the time* (4). A mean score was calculated with higher scores indicating greater academic engagement ($\alpha = 0.86$). The scale has demonstrated adequate reliability and validity with U.S. ethn racially diverse youth (Smalls and Cooper, 2012) and support for its use in Spanish has emerged in prior work (Núñez and León, 2019).

Depressive symptoms

Adolescents completed the 20-item Center for Epidemiologic Studies Depression Scale (CES-D; Radloff, 1977), which assessed symptoms associated with depression during the last week (e.g., "I felt depressed"). Responses ranged from *Rarely or none of the time* (0) to *Most of the time* (3). Scores were calculated such that higher scores indicated a greater number of depressive symptoms ($\alpha = 0.89$). The scale has demonstrated adequate reliability and validity with U.S. ethn racially diverse Latino youth (Umaña-Taylor and Updegraff, 2007), and support for its use in Spanish has emerged in prior work (Zeiders et al., 2015).

Self-Esteem

Adolescents completed the 10-item Rosenberg Self-Esteem Scale (Rosenberg, 1979), which assessed their self-esteem in the last year (e.g., "On the whole I am satisfied with myself"). Responses ranged from *Strongly disagree* (1) to *Strongly agree* (4). Higher scores reflected higher self-esteem ($\alpha = 0.87$). The scale has demonstrated adequate reliability and validity with U.S. ethn racially diverse youth (Umaña-Taylor et al., 2004), and its use in Spanish has been validated in prior research (Martín-Albo et al., 2007).

Other group orientation

Adolescents completed the 6-item Other Group Orientation Subscale of the Multigroup Ethnic Identity Measure (Phinney, 1992), which assessed the degree to which individuals felt positively about and interacted with ethn racial groups other than their own (e.g., "I like meeting and getting to know people from ethn racial groups other than my own"). Responses ranged from *Strongly disagree* (1) to *Strongly agree* (4).

(4). Higher scores indicated a more positive orientation toward engaging with out-group members ($\alpha = 0.74$). The subscale has demonstrated adequate reliability and validity with U.S. ethn racially diverse Latino youth (Guzmán et al., 2005).

Covariates

Control variables included adolescents' age (calculated using participants' parent-reported birthdate and survey completion date), gender (0 = male; 1 = female), and five dummy coded variables for school site (0 = "School 1," site with the highest number of participants in this sample as the reference group, $n = 163$; 1 = comparison school site, one each for Schools 2–6).

Analytic approach

Descriptive statistics are in Table 1. We tested a chain sequential mediation model that included: ERI exploration as a predictor of ERI resolution, ERI resolution as a predictor of global identity cohesion, and global identity cohesion as a predictor of the primary outcomes (i.e., GPA, academic engagement, self-esteem, depressive symptoms, other group orientation), controlling for participants' gender, age, and five dummy codes for school site (with the largest school as the reference group, $n = 163$). A bias-corrected bootstrap mediation path model with 10,000 bootstrap sample draws was tested in Mplus 8.4 to obtain unbiased confidence intervals for the indirect effects (MacKinnon et al., 2004). Covariances were estimated for all variables. We considered multiple fit indices in evaluating good model fit, including a non-significant χ^2 test of model fit, comparative fit index (CFI) ≥ 0.95 , root mean square error of approximation (RMSEA) ≤ 0.06 , and standardized root mean square residual (SRMR) ≤ 0.08 (Hu and Bentler, 1999). Estimates of specific indirect effects were examined, with 95% confidence intervals (CI) not including zero indicating statistically significant results. As a sensitivity analysis, we tested an alternate model that reversed the chain sequential ordering of ERI exploration and resolution to provide preliminary information regarding the potential direction of mediation effects.

Results

The fit of the hypothesized chain sequential mediation model was good: $\chi^2(14) = 17.457$, $p = .233$; CFI = 0.996; RMSEA = 0.023, 90% CI [0.000, 0.053]; SRMR = 0.019 (Fig. 1). R^2 effect sizes indicated the model predictors collectively accounted for 23.6% of the variance in GPA ($p < .001$), 20.1% for academic engagement ($p < .001$), 19.5% for depressive symptoms ($p < .001$), 30.4% for self-esteem ($p < .001$), and 8.5% for other group orientation ($p = .002$). Supporting the theorized

Table 1
Descriptive statistics and bivariate correlations among study variables.

	1.	2.	3.	4.	5.	6.	7.	8.	9.	10.
1. Self-Esteem	–									
2. Depressive Symptoms	–0.64***	–								
3. Grade Point Average	–0.05	0.04	–							
4. Academic Engagement	0.21***	–0.12**	0.02	–						
5. Other Group Orientation	0.08	0.03	0.08	0.20***	–					
6. Ethnic-Racial Identity Exploration	0.15**	–0.03	–0.06	0.21***	0.18***	–				
7. Ethnic-Racial Identity Resolution	0.19***	–0.12*	–0.03	0.20***	0.18***	0.48***	–			
8. Global Identity Cohesion	0.52***	–0.36***	–0.10*	0.31***	0.11*	0.11*	0.28***	–		
9. Age	0.05	–0.01	–0.12*	0.03	–0.05	0.04	–0.05	0.01	–	
10. Gender	–0.12*	0.20***	0.07	0.08	0.06	–0.09	–0.10*	0.02	0.02	–
<i>M</i>	3.18	0.95	3.58	2.87	3.21	1.80	2.59	3.91	15.90	.47 ^a
<i>SD</i>	0.59	0.55	0.42	0.61	0.48	0.72	0.83	0.49	1.15	–

Note. $N = 462$. Gender: 1 = female, 0 = male. ^aDichotomous variable presented as proportion of sample.

* $p < .05$.

** $p < .01$.

*** $p < .001$.

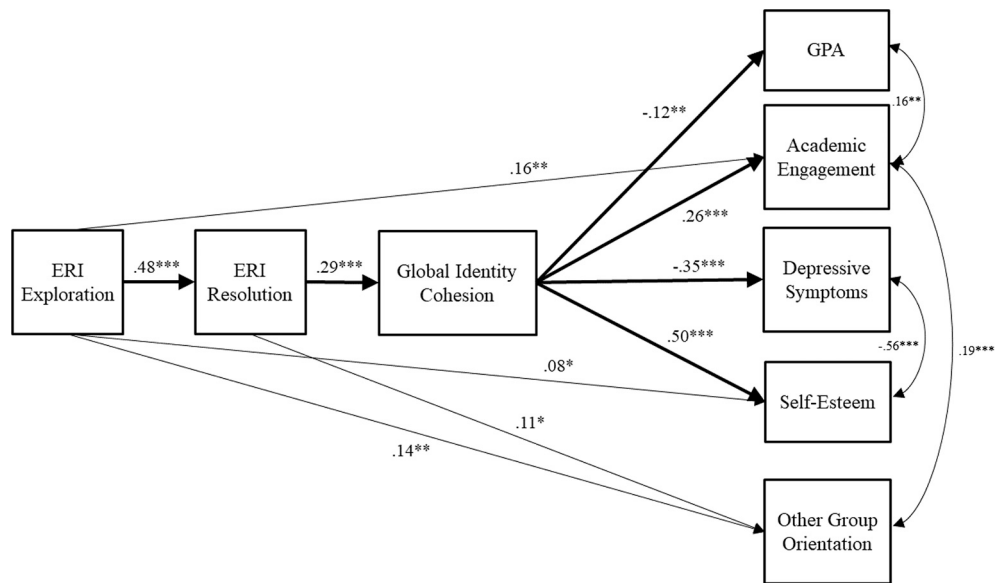


Fig. 1. Mediation path analysis model of ethnic-racial identity exploration, resolution, global identity cohesion, and adjustment. Note. $N = 462$. Model fit: $\chi^2(14) = 17.457$, $p = .233$; RMSEA = .023 (90% CI: .000, .053); CFI = .996; SRMR = .019. ERI = ethnic-racial identity. GPA = grade point average. Estimates are standardized. Bold paths represent significant indirect effects from ERI Exploration to outcome variables via ERI Resolution and Global Identity Cohesion. Non-significant paths ($p > .05$) omitted from figure for ease of presentation. All exogenous variables and outcomes allowed to covary. Control variables for all outcomes included gender (1 = female, 0 = male), age, and five dummy codes to control for school site (0 = "School 1," site with the highest number of participants in this sample as the reference group, $n = 163$; 1 = comparison school site, one each for Schools 2-6). See Results section for information about statistical significance of associations between control variables and outcomes. * $p < .05$. ** $p < .01$. *** $p < .001$.

model, path estimates for direct effects indicated that ERI exploration was positively associated with ERI resolution, and ERI resolution was positively associated with global identity cohesion (Fig. 1). As expected, global identity cohesion was positively associated with academic engagement and self-esteem and negatively associated with depressive symptoms. In an unexpected result, and consistent with the bivariate correlations, global identity cohesion was negatively associated with GPA. Regarding other significant direct effects, ERI exploration was also positively associated with academic engagement, self-esteem, and other group orientation, and ERI resolution was also positively related to other group orientation, over and above the variance in these outcomes accounted for by global identity cohesion. With respect to covariates that were significant ($p < .05$), age was negatively associated with GPA; relative to boys, girls reported higher levels of depressive symptoms, lower levels of self-esteem, and higher levels of other group orientation; relative to School 1 students (i.e., reference group with highest number of participants in the sample; $n = 163$), School 2 students ($n = 31$) had higher GPA, lower academic engagement, and higher other group orientation; School 3 students ($n = 107$) had higher GPA, lower academic engagement, and higher self-esteem; School 5 students ($n = 29$) had lower depressive symptoms and higher self-esteem; and School 6 students ($n = 79$) had lower GPA. All of these mean level differences were controlled for in our hypothesized model.

Regarding indirect effects, as hypothesized, ERI exploration was positively associated with ERI resolution, which, in turn, was positively associated with global identity cohesion; global identity cohesion, in turn, was positively associated with academic engagement (unstandardized estimate of indirect effect = 0.029, 95% CI [0.016, 0.046]), positively associated with self-esteem (indirect effect = 0.055, 95% CI [0.033, 0.080]), and negatively associated with depressive symptoms (indirect effect = -0.037, 95% CI [-0.055, -0.021]). This same mediational path was also significant for GPA, but contrary to hypotheses, ERI exploration was negatively associated with GPA via its associations with ERI resolution and global identity cohesion (indirect effect = -0.010, 95% CI [-0.018, -0.003]). Finally, though there was a significant direct effect, the mediational path from ERI exploration to other

group orientation via ERI resolution and global identity cohesion was not significant (indirect effect = 0.007, 95% CI [-0.002, 0.017]).

Alternate model

An alternate model using ERI resolution as a predictor of ERI exploration (rather than the reverse) had good fit: $\chi^2(14) = 19.903$, $p = .133$; CFI = 0.993; RMSEA = 0.030, 90% CI [0.000, 0.058]; SRMR = 0.019 (Fig. 2). Regarding indirect effects, ERI resolution was not significantly associated with GPA (unstandardized estimate of indirect effect = 0.001, 95% CI [-0.002, 0.004]), academic engagement (indirect effect = -0.002, 95% CI [-0.011, 0.007]), self-esteem (indirect effect = -0.004, 95% CI [-0.020, 0.012]), depressive symptoms (indirect effect = 0.003, 95% CI [-0.008, 0.014]), or other group orientation (indirect effect = -0.0004, 95% CI [-0.003, 0.002]) via associations with ERI exploration and global identity cohesion. Rather, replicating findings from the latter part of our hypothesized model (Fig. 1), indirect effects in this alternate model were significant for ERI resolution to GPA (unstandardized estimate of indirect effect = -0.018, 95% CI [-0.033, -0.006]), academic engagement (indirect effect = 0.053, 95% CI [0.030, 0.083]), self-esteem (indirect effect = 0.101, 95% CI [0.062, 0.144]), and depressive symptoms (indirect effect = -0.067, 95% CI [-0.099, -0.039]) via global identity cohesion (Fig. 2). Findings regarding the covariates in the alternate model demonstrated the same pattern of results as described above. Taken together, findings provided stronger support for the theorized process of ERI exploration predicting multiple indicators of adjustment via ERI resolution and global identity cohesion (Fig. 1), compared to results from an alternate model of ERI resolution predicting adjustment via ERI exploration and global identity cohesion (Fig. 2).

Discussion

The findings from the current study increase our understanding of the associations between Colombian adolescents' ERI development and their adjustment and begin to fill a significant gap in the literature.

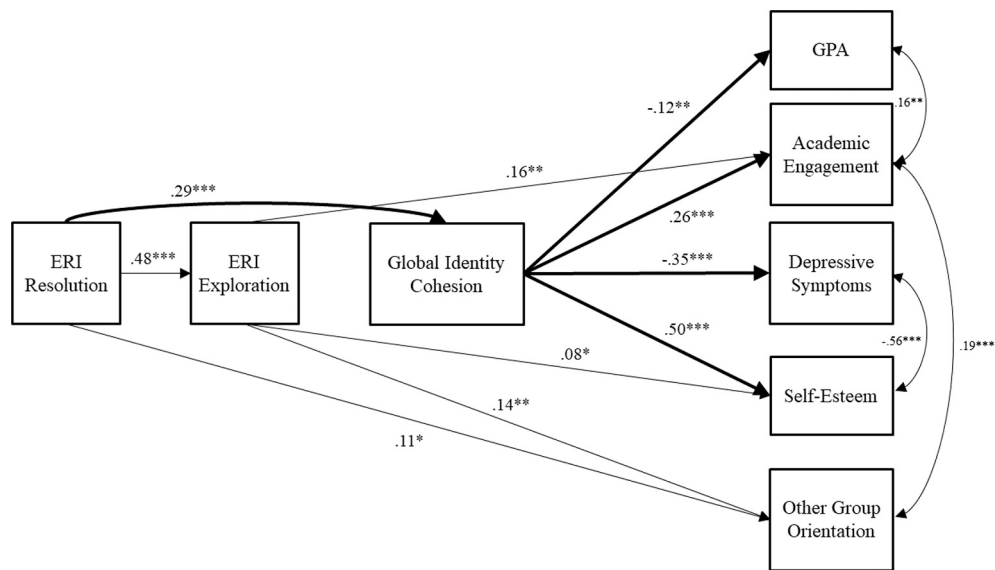


Fig. 2. Alternate mediation path analysis model.

$N = 462$. Model fit: $\chi^2(14) = 19.903$, $p = .133$; RMSEA = 0.030 (90% CI: 0.000, 0.058); CFI = 0.993; SRMR = 0.019.

ERI = ethnic-racial identity. GPA = grade point average. Estimates are standardized. Bold paths represent significant indirect effects from ERI Resolution to outcome variables via Global Identity Cohesion. Non-significant paths ($p > .05$) omitted from figure for ease of presentation. All exogenous variables and outcomes allowed to covary. Control variables for all outcomes included gender (1 = female, 0 = male), age, and five dummy codes to control for school site (0 = "School 1," site with the highest number of participants in this sample as the reference group, $n = 163$; 1 = comparison school site, one each for Schools 2–6). See Results section for information about statistical significance of associations between control variables and outcomes.

* $p < .05$. ** $p < .01$. *** $p < .001$.

Despite the absence of theoretical or empirical attention to Colombian adolescents' ERI development in the literature, prior research suggests that marginalization based on this social identity is salient in the lives of Colombians (e.g., Frydenberg et al., 2001; Hristov, 2009; Pereira and Telles, 2014), and the findings from the current investigation further suggest that ERI development may also be a salient developmental construct for Colombian youth – consistent with emerging evidence among minoritized youth in Chile and México (Briceño-Olivera et al., 2020; Guitart et al., 2011).

In line with an ERI developmental cascade model (i.e., Umaña-Taylor, 2018), we found support for a sequential mediation model in which higher ERI exploration was associated with higher resolution; higher ERI resolution was associated with greater global identity cohesion and, in turn, higher academic engagement, lower depressive symptoms, and higher self-esteem. Although the findings were derived from a cross-sectional design, the hypothesized direction of effects were firmly grounded in well-established theories and supported by prior longitudinal and experimental work with other populations (e.g., Umaña-Taylor, 2018) – features that can make the use of cross-sectional data to examine mediation informative for understanding temporal processes (Shrout, 2011). Furthermore, the results from an alternate model provided compelling support for the theoretical notion that the mediational mechanism that helps to explain the association between ERI and youth adjustment is one in which ERI exploration informs resolution, and through this sequential process is associated with global identity cohesion and other indices of adolescent adjustment (e.g., mental health, academic engagement). These findings provide preliminary support with a sample of ethn racially and socioeconomically diverse youth in Colombia for a universal developmental process model of ERI development that contributes to positive adjustment via global identity cohesion (Umaña-Taylor, 2018). Although context introduces important variability into human development processes, there are aspects of all developmental processes that are universal and thus replicate across contexts (Bornstein, 2017). The sociohistorical salience of race and ethnicity in Colombia, coupled with evidence of individuals' experiences with ethnoracial marginalization in this context, likely increases

Colombian youths' awareness of issues of race and ethnicity. Furthermore, this makes ERI formation a developmentally salient construct during adolescence when youth are actively engaged in defining themselves and considering their futures. Our findings are consistent with these notions and offer novel contributions to theory regarding the potential universally promotive role of ERI development during adolescence in countries characterized by ethnoracial inequities. An important next step will be to expand our understanding of the nuanced nature of ethnoracial discrimination and ERI development by examining how phenotypic variability is linked with Colombian adolescents' experiences with discrimination and the manner in which ERI can buffer the negative effects of such experiences on youth's adjustment. Furthermore, research with more rigorous designs (e.g., longitudinal, experimental) and an expanded sampling frame that includes other cities or regions in Colombia to examine the replicability and robust nature of these associations will be important; nevertheless, these initial findings are encouraging for further investment in this work with youth in Colombia.

Although we found much support for our theorized model, one notable unexpected finding pertained to direct associations between the ERI constructs and indices of adolescent adjustment. First, ERI exploration emerged as a *direct predictor* of both academic engagement and self-esteem. Specifically, adolescents who reported higher engagement in ERI exploration also tended to report being more academically engaged and to have higher self-esteem. As youth explore their ethnoracial background in greater depth, they can develop a deeper understanding and recognition of the strengths, history, cultural values, sense of community, and resilience associated with their ethnoracial group, which can contribute to building pride and self-confidence and also provide important counter narratives. Moreover, through exploration, youth can gain information that increases their awareness of social stratification systems, which can perhaps provide alternative explanations for instances of marginalization (e.g., due to external forces rather than personal attributes; Heberle et al., 2020; Neblett Jr. et al., 2012). It is also possible that this increased awareness may also reduce the cognitive energy spent on trying to make sense of discriminatory

experiences, perhaps enabling youth to sharpen their focus and be more academically engaged. Furthermore, increased ERI exploration may help youth form a more contextualized understanding of ethnoracial difference, rather than an essentialized understanding of ethnoracial difference, which can promote positive adjustment by helping youth navigate social group differences (Stephens et al., 2019). Of course, our conclusions are speculative, as the cross-sectional nature of our design also leaves open the possibility that those with higher self-esteem and who are more academically engaged would be more likely to engage in exploration of their ERI.

In addition, although we did not find support for our mediational hypothesis that ERI exploration and resolution would be associated with adolescents' positive attitudes toward outgroup members via the hypothesized mediational process, significant direct associations did emerge for each of these ERI constructs and adolescents' other group orientation. Specifically, as adolescents reported higher ERI exploration, they also tended to report more positive attitudes toward engaging with others who are ethnoracially different from them. Similarly, adolescents who reported higher levels of ERI resolution tended to also report having a more positive outgroup orientation. As adolescents increasingly explore their ERI and develop a deeper understanding of their own ethnoracial background, this may raise curiosity about others' backgrounds as adolescents are attempting to understand the self in context. Indeed, the current findings are consistent with prior research with U.S. adolescents (Sladek et al., 2020) and college students (Phinney et al., 2007), in which higher ERI exploration and resolution were associated with a more positive outgroup orientation. The greater sense of self-confidence that comes with feeling resolved, clear, and well-informed on the meaning of one's ethnoracial background for one's self-concept may enable youth to feel more comfortable connecting across ethnoracial differences. Indeed, as youth feel more confident and secure in their sense of self, theory suggests this enables them to engage in more productive and healthy relationships with others (Erikson, 1968). Generally, these findings provide preliminary evidence for the notion that as Colombian adolescents engage more deeply in the developmental process of considering their own ERI and the role that it plays in their lives, they also become more open to intergroup contact.

A final unexpected finding was that youth who reported higher global identity cohesion also tended to have lower grades. First, this counterintuitive finding raises questions about whether the construct of GPA may carry a different meaning in the Colombian cultural context compared to how it is typically conceptualized in the U.S. Although GPA is an important indicator of academic achievement in the U.S. and has significant practical implications such as for college admissions (e.g., Allensworth and Clark, 2020), in Colombia, GPA is not considered for college admissions and, instead, admission is based on a standardized high school exit test (i.e., SABER 11), sometimes in combination with interviews and university-specific entrance exams (Barrera-Osorio and Bayona-Rodríguez, 2019). Thus, GPA may not be as relevant as an indicator of positive adjustment for Colombian adolescents. In addition, the negative association between global identity cohesion and GPA could potentially reflect a lack of emphasis placed on seemingly objective indicators of achievement, such as grades, among youth who have a clearer and more confident sense of self. Such individuals may be less concerned with institutional evaluations of their achievements and thus place relatively less emphasis on competing for higher grades and relatively more emphasis on developing their peer relationships or other social aspects of schooling, which could potentially result in lower marks in school. Yet another possibility is that students who have been highly academically engaged and have struggled with their grades despite their high academic engagement have had to engage in deeper exploration and reflection on their global identity, which has led to a greater sense of identity synthesis and less identity confusion. In short, more work is needed to better understand the relevance of GPA as an indicator of adolescent adjustment in Colombia and, particularly, to understand why it would be negatively associated with adolescents'

sense of clarity and understanding about themselves and their global identity.

Despite its many strengths and important contributions to nascent literature, the current study is not without limitations. Although findings provide preliminary support for a developmental cascade model, a significant limitation of this study is that the findings are based on cross-sectional and non-experimental data. Although mediation analyses using cross-sectional data can be informative about temporal processes when the hypothesized direction of effects are grounded in well-founded theoretical arguments (Shrout, 2011), as was the case in the current study, this aspect of our study design nevertheless limits our ability to conclude direction of effects. As such, it will be important for future research to test these ideas with prospective longitudinal designs and, ideally, experimental methods that can be used to infer causality (e.g., Schachner et al., 2024; Umaña-Taylor et al., 2018). Barring such work, the current findings indicate interesting relations among these constructs and identify important areas for future research. Relatedly, although the current study included adolescents from six unique school communities across a large city in Colombia, and selected schools that were representative of the six socioeconomic status groups in the country, the sample is nevertheless a community-based convenience sample. Replication studies will be important with youth from other school communities within Medellín and in other regions of Colombia.

Conclusion

Developing an understanding of the self involves exploring all salient aspects of one's identity. When it comes to social identities that are salient in a society, the exploration and resolution of one's sense of self with respect to that particular social identity is important for developing a cohesive sense of self. This, in turn, translates into other psychosocial benefits (e.g., academic success, psychological adjustment) that are important for determining success in society and, ultimately, shaping one's life chances. Regardless of whether the broader society explicitly acknowledges the significance that is placed on a given social identity such as race, individuals' lived experiences as informed by their membership in particular groups that experience differential treatment based on this social identity will play a role in how they view themselves and the implications this has for their adjustment. Grounded in notions from developmental and social identity theories, ERI theory suggests that an explicit focus on exploring this aspect of the self and gaining a sense of clarity regarding the role that this social identity will play in one's life will be important for individuals' positive adaptation in a society where race and ethnicity are salient social constructs. Empirical support for these theoretical notions has significant implications for practice, as it informs preventive intervention efforts that can have important downstream impacts that contribute to reducing ethnoracial inequities. The findings from the current study provide preliminary support for these theoretical notions and suggest that ERI-focused prevention efforts are worthy of further study with Colombian youth.

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Compliance with ethical standards

All procedures performed in this study that involved human

participants were in accordance with the ethical standards of the institutional research committee at the Universidad de San Buenaventura and with the 1964 Helsinki Declaration and its later amendments or comparable ethical standards.

Informed consent

Informed consent was obtained from all individual adult participants included in the study; assent was obtained from children. We have no known conflicts of interest to disclose.

CRediT authorship contribution statement

Adriana J. Umaña-Taylor: Conceptualization, Methodology, Formal analysis, Investigation, Resources, Data curation, Writing – original draft, Supervision, Project administration, Funding acquisition. **Michael R. Sladek:** Formal analysis, Software, Visualization, Data curation, Writing – original draft, Writing – review & editing. **M. Dalal Safa:** Data curation, Software, Writing – review & editing. **Liliana M. Uribe Tirado:** Investigation, Resources, Project administration, Funding acquisition, Writing – review & editing. **Luz Magnolia Tilano Vega:** Investigation, Project administration, Writing – review & editing.

Declaration of competing interest

The authors have no known conflicts of interest to disclose.

Data availability

The authors do not have permission to share data.

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